كلية تكنولوجيا الصناعة والطاقة

Information Technology Department
2rd Year

جامعة برج العرب التكنولوجية

Course: Semester:2rdterm 2025/2026	Lecturers: Dr. Radwa Rady Dr. Ghada Fathy
Java Programming **Java Project**	

1- Student Management System

Project Description

Design a **Student Management System** that allows the user to manage student information.

In this project, build a **Student Management System** that manages:

- Students
- Subjects (Courses)
- Grades for each subject
- GPA calculation

The system should allow users to **add students, assign subjects to students, record grades, and calculate the GPA** for each student.

Required Features

The system should allow the user to:

Example menu:

------ Student Management System ------

```
1. Add Student
2. Add Subject to Student
3. Display Student Information
4. Calculate Student GPA
5. Display All Students
6. Exit
```

Use a **menu-driven program** that keeps running until the user chooses to exit. Using GUI (**Optional with Bonus**)

Required Classes

You must implement at least three classes**.**

1- Student Class

Represents a student in the system.

Attributes:

```
Student
 ---------
 id
 name
 major
 ArrayList<Subject> subjects
```

Methods:

```
addSubject()
calculateGPA()
displayStudentInfo()
```

2- Subject Class

Represents a subject taken by a student.

Attributes:

```
Subject
 ---------
 subjectName
 creditHours
 grade
```

3- StudentManagementSystem Class

This class manages the entire system.

Attributes:

```
StudentManagementSystem
------------------------
ArrayList<Student> students
```

Methods

```
Add new students()
Search for students()
Add subjects to students()
Display student information()
Calculate GPA()
```

Note: GPA Calculation

Students must implement **GPA calculation based on credit hours**.

Formula

$$GPA = \sum(grade \times creditHours) / \sum(creditHours)$$

Example

Subject	Credit	Grade
Java	3	90
Database	3	80
Data Structures	4	95

Calculation:

(90×3 + 80×3 + 95×4) / (3+3+4)

= (270 + 240 + 380) / 10
= 890 / 10
= 89.0

Example Interaction

===== Student Management System =====

```
1. Add Student
2. Add Subject to Student
3. Display Student Information
4. Calculate Student GPA
5. Display All Students
6. Exit
```

Feature 1: Add Student

```
Enter your choice: 1
Enter Student ID: 101
Enter Name: Ahmed Ali
Enter Major: Computer Science
"Student is added successfully"
```

Feature 2: Add Subject to Student

```
Enter your choice: 2
Enter Student ID: 101
```

```
Enter Subject Name: Java Programming
Enter Credit Hours: 3
Enter Grade: 92
"Subject is added to that student".
```

Feature 3: Display student's info:

```
Enter your choice: 3

Student ID: 101
Name: Ahmed Ali
Major: Computer Science

Subjects:
Java Programming - Grade: 92 - Credit Hours: 3
Database Systems - Grade: 85 - Credit Hours: 3
Data Structures - Grade: 90 - Credit Hours: 4
-----------------------------------------------------------------
```

Feature 4: Calculate GPA

```
Enter your choice: 4
Enter student ID: 104

Student: Ahmed Ali
GPA: 89.4
```

Feature 5: Display all students

```
Enter your choice: 5

Students in the system:

1. Ahmed Ali
2. Sara Mohamed
3. Omar Khaled
```

For Big Team (8 Members)

===== Student Management System =====

```
 1. Add Student
 2. Add Subject to Student
 3. Display Student Information
 4. Calculate Student GPA
 5. Display All Students
 6. Remove Student
 7. Remove Subject
 8. Show student with highest GPA
 9. Convert numeric grade to letter grade
 10.  Exit
```

```
List of Students:
1. Ahmed Ali (ID: 101)
2. Sara Mohamed (ID: 102)
3. Omar Khaled (ID: 103)

-----------------------------------------------------------------
```

Feature 6: Remove Student

```
Enter your choice: 6

Enter Student ID to remove: 102

Student removed successfully.
```

After Removal

```
Students in system:
1. Ahmed Ali
2. Omar Khaled
```

Testing Case

```
Enter Student ID to remove: 999

Error: Student not found.
-----------------------------------------------------------------------------------------
```

Feature 7: Remove Subject

```
Enter your choice: 7

Enter Student ID: 101

Subjects:
1. Java Programming
2. Database Systems
3. Data Structures

Enter subject number to remove: 2

Subject removed successfully.
```

After Removal

```
Student ID: 101
Name: Ahmed Ali

Subjects:
Java Programming - Grade: 92
Data Structures - Grade: 90
```

Testing Case:

```
Enter subject number to remove: 5

Error: Invalid subject number.
-----------------------------------------------------------------
```

Feature 8: Calculate Highest GPA

```
Enter your choice: 8

Calculating highest GPA...

Ahmed Ali → GPA = 3.2
Sara Mohamed → GPA = 3.8
Omar Khaled → GPA = 3.5
```

Output

```
Student with Highest GPA:

Name: Sara Mohamed
ID: 102
GPA: 3.8
----------------------------------------------------------------
```

Feature 9: Convert Numeric Grade to Letter Grade

```
Enter your choice: 9

Enter numeric grade: 85

Letter Grade: B
```

Note:

Grade Conversion Rule

```
90 - 100 → A
80 - 89  → B
70 - 79  → C
60 - 69  → D
< 60     → F
```

Instructions

You have check the input from user is correct

1- Input Validation

Your program should check for invalid input such as:

- Negative grades
- Grades greater than 100
- Empty student names
- Student not found when searching

Example:

```
Invalid grade. Please enter a value between 0 and 100.
```

2- Code Organization

Your project should be organized clearly:

Example structure:

```
Student.java
Subject.java
StudentManagementSystem.java
Main.java
```

Each class should be in a **separate file**.

3- Your code must include:

- Proper indentation.
- Meaningful variable names.
- Comments explaining important parts of the code.

4- Testing the Program

Before submitting, test your program with different cases:

Test examples:

- Add multiple students
- Add several subjects for each student
- Calculate GPA correctly
- Search for a student that does not exist
- Try invalid grades

Make sure the program **does not crash**.

You will be asked to **explain your code during evaluation**.

Start early and build the project **step by step** ☺ ☺.

Good Luck